Complete Website Architecture & Component Audit

A source-backed review of the 2026 platform rebuild, registration system, privacy boundary, visual system, and deployment readiness.

Audit date	Branch	Health score	Release state
05 November 2026, 15:55 IST	feature/2026-platform-rebuild	92 / 100	Code complete; private preview blocked by expired Cloudflare authentication

Confidentiality boundary

This document describes the implementation only. It does not reproduce private school correspondence, signatures, partner records, participant data, or unapproved achievements.

A lightweight learning platform with a real operations layer.

Area	Status	Evidence
Frontend	PASS	Responsive React Router interface; no horizontal overflow at 390, 768, or 1440 pixels.
Registration	PASS	D1 persistence, unique public IDs, idempotency, guardian workflow, queue-backed email jobs, Excel and CSV exports.
Accessibility	PASS	Eight Playwright tests; axe critical and serious checks include colour contrast.
Security	PASS WITH CONFIG	No committed secrets found. CSP, HSTS, secure sessions, Turnstile, origin checks, and hashed rate-limit identifiers implemented.
SEO	PASS	Unique metadata, canonical URLs, sitemap, robots policy, and preserved Cloudflare Pages production origin.
Deployment	BLOCKED EXTERNALLY	Cloudflare CLI token expired. No production DNS or live site was modified.

Architecture

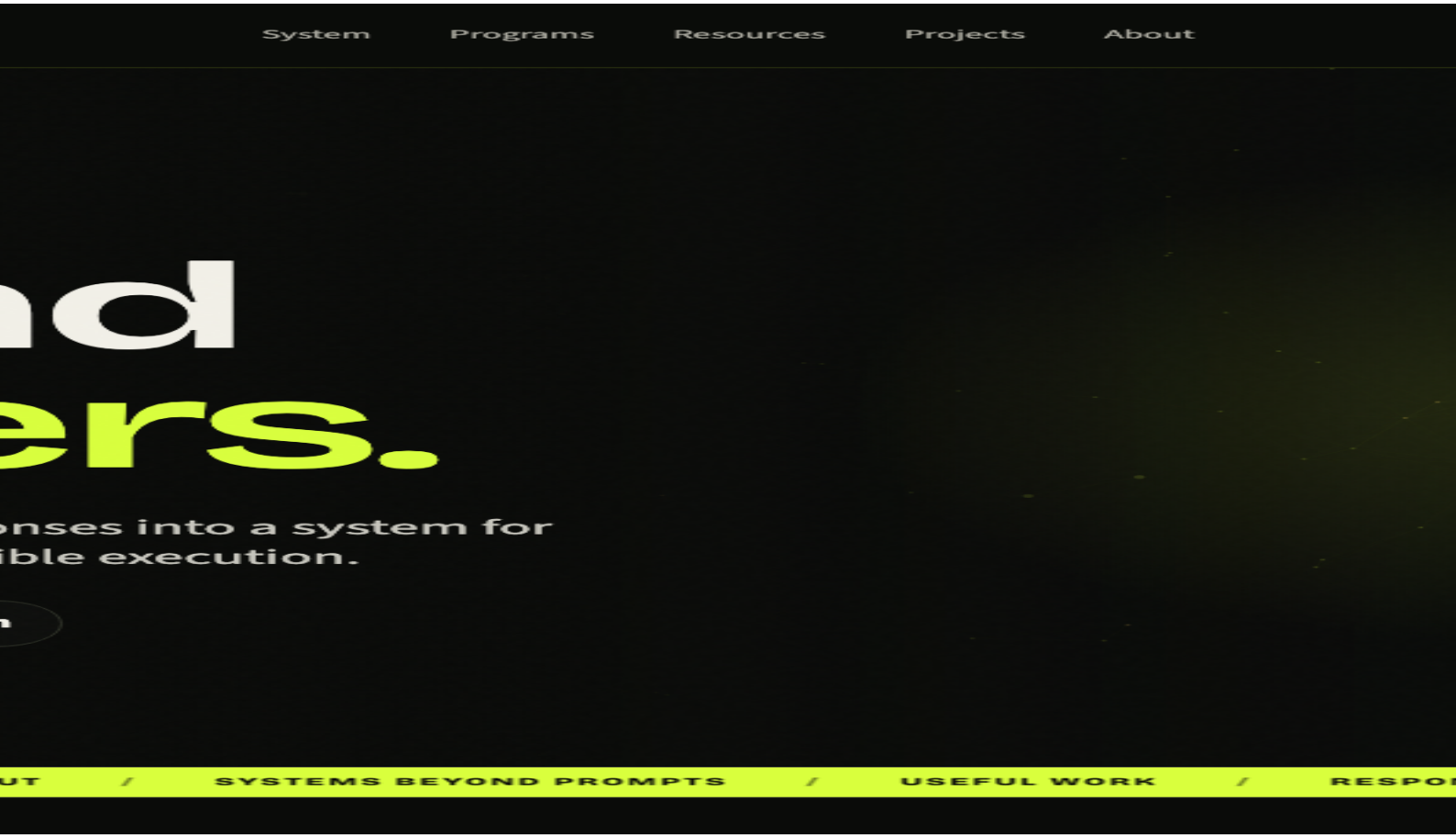
React Router 8 and TypeScript render on Cloudflare Workers. D1 stores registrations and consent records. Cloudflare Queues decouple email delivery. Resend is provider-ready but suppressed when no key is present. Public pages receive canonical metadata; operational routes are noindex and access controlled.

Hosting decision

Cloudflare was selected over Netlify for the rebuild because the current indexed production origin is already on Cloudflare Pages and the registration requirements benefit from a single D1, Queues, Turnstile, and Worker boundary. Netlify remains a documented static fallback, but it would add database and queue vendors for the same workflow.

Systems in Motion

Selected after evaluating three complete directions: Systems in Motion, Human to Human, and Living Lab.



Decision	Implementation
Concept	Beyond Answers is expressed as movement from prompts to workflows, systems, and outcomes.
Typography	Bricolage Grotesque for editorial hierarchy; Source Sans 3 for readable body copy; both served locally.
Palette	Near-black, paper, and acid-lime. No generic AI gradient or stock illustration layer.
Motion	One lightweight canvas signal field with reduced-motion fallback. No scroll hijacking or autoplay media.
Visual QA	Three screenshot rounds. Final score: 47 / 50. Initial hidden reveal gaps and registration spacing were fixed.

Component map and content audit

Component and path	Rendered content	Health and action
SiteHeader / app/components/site-shell.tsx	Lumenet Labs; System; Programs; Resources; Project; About; Register	Keyboard-accessible navigation. Accessible name now matches visible brand text.
HomeHero / app/routes/home.tsx	Beyond answers. Register for 25 November. See the system.	Canvas is decorative and reduced-motion safe. CTA appears early on mobile.
EvolutionSequence / app/routes/home.tsx	Prompts; Workflows; Systems; Outcomes	Semantic list-like sequence with strong scanning order and no dependency on animation.
Outputs / app/routes/home.tsx	Research briefs; Study systems; Project workflows; Responsible use	No invented metrics or partner claims.
EventPanel / app/routes/home.tsx	Evolve with AI; 25 / 11 / 26	Only confirmed date is public. Time, venue, delivery format, and fee remain explicitly unconfirmed.
SiteFooter / app/components/site-shell.tsx	Privacy; Terms; Contact; Instagram	Minimal footer with direct policy access.

Minimal data, explicit consent, operational clarity



Block	Exact purpose	Implementation note
Event facts	25 November 2026; time and venue to be confirmed; guardian approval under 18	Prevents invented operational certainty.
Participant fields	Name, email, school or community, learner category, age band, optional interest	Zod limits lengths and enum values server-side.
Consent	Operational communication, privacy acknowledgement, optional marketing	Marketing is separate and unchecked by default.
Abuse controls	Honeypot, Turnstile, same-origin checks, salted IP hash rate limit	Raw IP addresses are not stored in registration rows.
Persistence	Unique public ID, idempotency key, active registration uniqueness	D1 batch writes registration, guardian consent, and email job before queue publication.
Operations	Authenticated dashboard, Excel and CSV export	Exports neutralise spreadsheet formulas and use no-store responses.

Controls implemented before preview

Control	Result
Secrets	No current secret pattern, private key, personal email, or tracked real environment file found.
Sessions	Eight-hour HttpOnly, Secure, SameSite Strict organizer cookie with HMAC verification.
Authorization	Organizer dashboard and export require a valid signed session.
Guardian consent	One-time random token stored only as SHA-256 hash; expiry enforced; successful completion no longer reuses the token.
Email safety	Participant content is HTML-escaped; preview email sends are suppressed without a provider key.
Headers	CSP, HSTS, frame denial, content-type protection, strict referrer policy, and restricted browser permissions.
Preview indexability	Non-production Worker responses include X-Robots-Tag: noindex, nofollow.
Retention	Documented schedule separates operational records, consent evidence, optional marketing consent, and diagnostics.

Privacy boundary

The public project note states that Ekya is in progress. It does not publish private CAS correspondence, signatures, partner records, participant data, unapproved logos, testimonials, or impact claims.

Tests run and passed

Gate	Result	Evidence
TypeScript	PASS	Cloudflare type generation, React Router type generation, and tsc project build.
Unit tests	PASS	2 tests for CSV and XLSX formula neutralisation and workbook output.
Browser tests	PASS	8 Playwright tests for axe, mobile overflow, registration labels, guardian completion, and organizer redirect.
Accessibility	PASS	Axe critical and serious checks with colour contrast enabled on home, registration, privacy, and resources.
Production build	PASS	Client and Worker builds complete. Home route is 2.49 KB gzipped; shared client runtime is 69.76 KB gzipped.
Dependency audit	PASS	npm audit reports zero vulnerabilities.
Cloudflare package	PASS	Wrangler dry run reports approximately 219 KB gzipped upload before cached assets.
Registration smoke	PASS	A real local POST returned HTTP 302 with a unique public ID and confirmed D1 record.
Visual review	PASS	Final screenshots at 390, 768, and 1440 pixels show no overflow, clipping, or hidden sections.

Top three next steps

1. Reauthenticate Wrangler, create the preview D1 database and queues, set secrets, run remote migrations, and deploy the noindex preview.

2. Configure a verified Resend sender only after communication copy, school safeguarding requirements, and data controller responsibilities are approved.

3. Complete stakeholder review of time, venue, format, fee, facilitators, capacity, consent wording, and publication approval before any production DNS or indexed-site change.